



CEU – Catalytic Reactors

PRODUCT CODE: **CEU**

Categories: CE Series - Chemical Engineering, CEU - Catalytic Reactors, Chemical, Educational

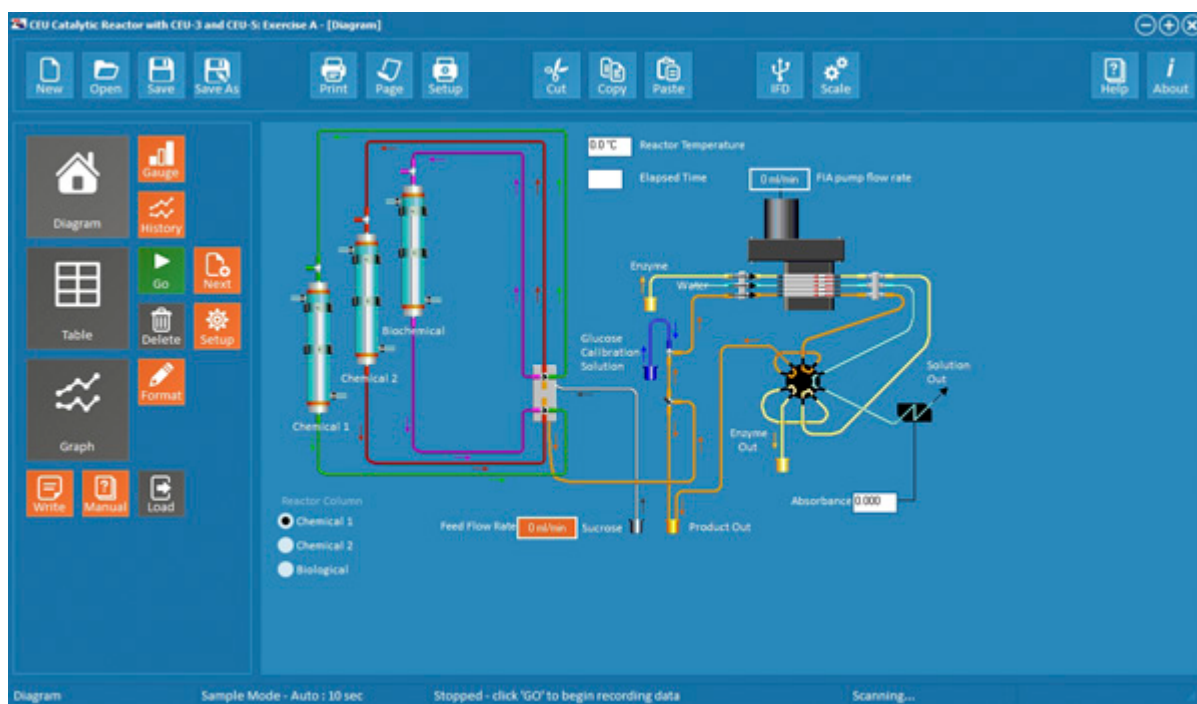
Tags: Biological Catalysis, Catalytic Reactors, CE Series, CEU, Chemical And Biological Catalytic Reactors, Chemical Catalyst, Chemical Engineering Basic Process Principles, Colorimetric, Enzyme Invertase, Enzymic Reactor, Flow Injection Analysis, Fructose, Glucose, Optical Absorbance Sensor, Optical Sensor, Packed Bed Catalytic Reactors, Reactor Columns, Reactor Kinetics, Sucrose, Sugar Inversion Reaction, Variable-Speed Peristaltic Feed Pump



DESCRIPTION

The CEU utilises the sugar inversion reaction (sucrose $\rightarrow$ glucose + fructose) to study the performance of packed bed chemical and biological catalytic reactors.

A colorimetric assay is used to determine the degree of conversion using an optical sensor. Assays may be automated using optional flow injection analysis.



TECHNICAL SPECIFICATIONS

FEATURES & BENEFITS

Experimental content

- Understanding the principles of packed bed catalytic reactors
- Mass balancing
- Examination of steady and unsteady state catalysis
- Comparison of chemical and biological (enzymic) catalysis
- Flow characterisation in a packed bed
- Understanding the principles of flow injection analysis (FIA)

Ordering specification

- A benchtop unit comprising a vacuum-formed ABS plastic plinth with integral electrical console on to which is mounted the packed bed reactor columns, feed pump, optical sensor and optional flow injection analysis (FIA) system
- The unit is supplied as standard with two reactor columns for chemical catalysis
- An optional third column can be added to the system for biological (enzymic) catalysis
- A heated water supply to the column jackets enables automatic control of reaction temperature to a set point value
- Feed flow rate 0-15 ml/min
- The optional FIA pump can be adjusted to give flow rates up to 2.5 ml/min
- All electrical circuits are fitted with appropriate protection devices
- The console has two digital meters. The first, associated with the controller, shows the temperature of water supplied to the column jackets and the second shows the optical sensor reading which provides a measurement of product concentration. Corresponding signals are routed to the I/O port for connection to a PC

Requirements

Electricity Requirements:

CEU-A: 220-240V / 1ph / 50Hz / 5A

CEU-B: 120V / 1ph / 60Hz / 10A

CEU-G: 220-240V / 1ph / 60Hz / 5A

Software:

Requires the user to have a PC running Windows 7 or above with a USB port.

Consumables:

500ml of 2M Hydrochloric acid

500ml of 2M Calcium chloride

3kg Amberlite

10g Glucose

38g Sucrose

1kg Liquid Glucose (oxidase)



250mg Invertase
30g of Sodium alginate

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.44m³

Gross Weight: 36kg

Dimensions

Length: 1.00m

Width: 0.50m

Height: 0.60m

Order Codes

CEU-A: 220-240V / 1ph / 50Hz / 10A

CEU-B: 120V / 1ph / 60Hz / 15A

CEU-G: 220-240V / 1ph / 60Hz / 10A

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